Label Quality Defines the Performance Ceiling for Peptide-MHC Binding Prediction: A Six-Architecture Benchmark with Multi-Allele Generalization

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Abstract

Motivation: Peptide-MHC class I binding prediction underpins epitope discovery for vaccines and cancer immunotherapy, yet the relative contributions of architecture choice, label quality, and peptide encoding to benchmark performance remain poorly disentangled.

Results: We benchmarked six classifier architectures on 5,088 MHCflurry 2.2.0-labelled 9-mers targeting HLA-A*02:01, across three label sources (MHCflurry, deterministic PSSM, random synthetic), two encodings (BLOSUM62, ESM-2), and external validation against 49 IEDB epitopes. Label source accounted for $\sim$29 percentage points of accuracy variation (PSSM 94.8% vs. random 65.8%), exceeding the maximum inter-architecture range ($\sim$11 pp); variation concentrated exclusively in the weak-binder class (F1 0.723–0.925). A deep feed-forward network achieved 91.9% accuracy; BLOSUM62 generalized more reliably than ESM-2 under cross-validation (92.2% vs. 82.2%). IEDB validation yielded 93.9% sensitivity (ROC AUC 0.947). A multi-allele extension showed one rebalanced pan model matches per-allele specialists across five alleles (0.819 pooled sensitivity on a 15-allele panel), while non-canonical binders plateau at $\sim$0.65 sensitivity across sequence- and structure-based predictors.

Availability: Code, models, and data at github.com/woodhaha/peptide-mhc-benchmark.

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Supplementary information: Supplementary data are available at *Bioinformatics* online.

1 Introduction

MHC class I molecules present 8–11-mer peptides to CD8+ T cells, and accurate prediction of peptide–MHC-I binding underpins epitope discovery for vaccine design and cancer immunotherapy (Falk *et al.*, 1991; Lundegaard *et al.*, 2011). For HLA-A*02:01 — the most prevalent class I allele globally — binding is dominated by two primary anchor positions: position 2 (p2, hydrophobic: Leu/Met/Ile/Val) and position 9 (p9, C-terminal, Val/Leu preferred) (Ruppert *et al.*, 1993; Rammensee *et al.*, 1995). Computational predictors have evolved from position-specific scoring matrices to neural networks and pan-specific tools such as NetMHCpan (Reynisson *et al.*, 2020), MHCflurry (O'Donnell *et al.*, 2020), BigMHC (Albert *et al.*, 2023), and MUNIS (Chen *et al.*, 2025), increasingly using protein language model embeddings (Rives *et al.*, 2021).

Several gaps remain. Head-to-head architecture comparisons under fully controlled conditions (identical training data, encoding, evaluation) are uncommon. The impact of training-label quality on downstream performance is seldom quantified, despite evidence that $>$55% of IEDB entries are computationally rather than experimentally labelled (Preibisch *et al.*, 2026). And the translation of binding prediction to real-world epitope discovery is rarely demonstrated in a unified, reproducible pipeline.

We address these gaps through a systematic benchmark designed to isolate the effects of architecture choice, label quality, and encoding. Our contributions are: (i) quantification of label-quality effects across three labelling strategies, showing label provenance accounts for more performance variation than architecture choice; (ii) a controlled six-architecture comparison on a common dataset; (iii) evaluation of ESM-2 embeddings against BLOSUM62 with a mean-pooled ablation; (iv) external validation against 49 IEDB epitopes; (v) translational application through protein scanning and cancer mutation analysis; (vi) a docking-based structural hypothesis for non-canonical anchoring; and (vii) a multi-allele extension testing whether the recipe generalizes across common HLA-I alleles and how hard non-canonical binders are for sequence- and structure-based predictors alike.

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2 Methods

2.1 Training data and labels

100,000 random 9-mer peptides were generated by uniform sampling of the 20 standard amino acids. Each was labelled with MHCflurry 2.2.0 against HLA-A*02:01 and assigned to Strong Binder (SB, percentile rank $<0.5\%$), Weak Binder (WB, $0.5\% \leq$rank$<2.0\%$), or Non-Binder (NB, rank $\geq 2.0\%$) following the netMHCpan convention (Reynisson *et al.*, 2020). The resulting distribution (1.7% SB, 5.1% WB, 93.2% NB) was balanced by downsampling to 1,696 per class (n=5,088), partitioned 90/10 into training (n=4,579) and held-out test (n=509) sets with stratified sampling (seed 42). Two comparator labelling strategies were implemented: a PSSM encoding crystallographic A*02:01 anchor preferences, and random synthetic labels as a baseline control. An external benchmark of 49 experimentally validated IEDB epitopes plus 20 homopolymer negatives was curated (Vita *et al.*, 2019).

2.2 Peptide encoding

Each 9-mer was encoded with the BLOSUM62 matrix (Henikoff and Henikoff, 1992), min–max normalized, producing a 9×20 matrix flattened to 180 dimensions. ESM-2 embeddings (Rives *et al.*, 2021) were extracted at t6 (2,880-dim per-position) and t12 (4,320-dim) depths; a mean-pooled t6 variant (320-dim) served as an ablation control destroying positional information.

2.3 Model architectures

Six classifiers were implemented: (i) FFN (180→90→3, 49K params); (ii) Deep FFN (180→360→180→90→45→3 with batch normalisation and dropout, 152K params); (iii) CNN (Conv2D(32,3×3) + FFN body, 1.05M params); (iv) LSTM(64)→dense(32)→3, 24K params; (v) ResNet-style stem + 3 residual blocks, 325K params; (vi) Random Forest (100 trees). All networks used cross-entropy loss with Adam or RMSprop (lr 0.001), 150 epochs, batch size 50, 20% internal validation split, early stopping (patience 10), and learning-rate reduction on plateau. Five-fold stratified cross-validation was performed for the FFN. The held-out test set (509 peptides) was disjoint from training and used for all reported external metrics, satisfying the independent-test-set requirement.

2.4 Protein scanning, mutation analysis, and structure

Ten therapeutically relevant proteins were scanned with a sliding 9-mer window (stride 1; 3,536 windows). Sixteen recurrent hotspot mutations (7 p53, 9 KRAS) were analyzed by comparing wild-type and mutant binding scores; mutations were classified as Created/Enhanced/Destroyed/Unchanged. The KRAS G12V neoepitope YKLVVVGAV carries a non-canonical lysine at P2; we evaluated a predicted K-E63 salt bridge using comparative B-pocket geometry across eight HLA-A*02:01 crystal structures (PDB 1DUZ, 1AKJ, 1HHJ, 1JF1, 1QEW, 1QRN, 1S9W, 2GIT), energy minimisation (OpenMM, amber14 force field), and a Coulombic electrostatic estimate.

2.5 Feature extraction

Learned features were extracted from the penultimate dense layer via truncated models. Hierarchical clustering (Ward's method, Euclidean distance) was applied to representations of top epitope candidates; feature–binding-score correlation was assessed by Pearson's r with a 1,000-iteration permutation test.

2.6 Multi-allele extension

Experimentally confirmed class I binders were curated from IEDB for five common alleles (HLA-A*02:01, A*01:01, A*03:01, B*07:02, B*44:03), yielding n=14,488 positives (canonical 11,225; Type A 151; Type B 3,112). Type A (tolerated weak anchors) and Type B (energy-compensated non-canonical) were defined by whether the peptide violates the allele's canonical anchor motif at p2/p9. Protein-background negatives were sampled from natural proteomes. A rebalanced pan-specific DFFN (B*07:02 upsampled $4\times$) was trained on pooled data with five seeds (0, 1, 42, 123, 2024). MHCflurry 2.2.1 (affinity <500 nM) and MUNIS (Chen *et al.*, 2025) (structure-based) served as independent comparators on the n=3,112 Type B population plus canonical controls. A 15-allele panel (pan16) tested scaling with per-allele minimum sample (2,000) upsampling. To address label circularity, temporal holdout compared IEDB epitopes published after 2022 against pre-2021 (in-distribution) epitopes at sensitivity threshold 0.95.

3 Results

3.1 Training data and architecture comparison

MHCflurry labelling of 100,000 random 9-mers yielded 1.7% SB, 5.1% WB, 93.2% NB; after balancing, 5,088 peptides (1,696/class). The Deep FFN achieved the highest accuracy (91.9%, macro F1 0.921; Table 1), outperforming FFN (90.9%), CNN (90.0%), ResNet (84.3%), LSTM (83.3%), and Random Forest (81.1%). MHCflurry itself scored 100% on the test set it labelled, quantifying label circularity. The WB class was the most difficult (F1 0.752–0.880). Five-fold CV of the FFN confirmed stable generalisation ($89.6 \pm 0.8\%$).

The Deep FFN's advantage over more complex architectures reflects the positional nature of peptide–MHC binding: anchor positions p2/p9 dominate, and convolutional or recurrent mixing of positional information adds little.

3.2 Label quality dominates

Label source was the dominant determinant of performance (Table 2). PSSM labels yielded 94.8% accuracy; MHCflurry 91.9%; random synthetic 65.8%. The 2.9 pp PSSM–MHCflurry gap was concentrated in the WB class (F1 0.925 vs. 0.880); the random-label model's WB F1 of 0.000 reflects a model that never predicts the class.

3.3 Encoding

BLOSUM62 generalized stably under cross-validation ($92.2 \pm 0.4\%$). ESM-2 t6 per-position embeddings scored 93.3% on a single split but collapsed to $82.2 \pm 1.0\%$ under CV — the 2,880-dimensional features overfit the 4,579-sample training set. Mean-pooled t6 embeddings collapsed to 65.9%, confirming that positional information is essential.

3.4 External validation and translation

The Deep FFN achieved 93.9% sensitivity (46/49), 75.0% specificity, ROC AUC 0.947 against IEDB epitopes (Table 3). All three false negatives carried non-canonical p9 anchors; a zero-parameter anchor-checker baseline achieved sensitivity 1.0 and F1 0.85–1.0 on the canonical-dominated set, confirming that the canonical benchmark overstates generalisation. Protein scanning confirmed 10/11 known epitopes; mutation scanning identified epitope-altering events in p53 R248W and KRAS G12V (Table 4).

3.5 Structural hypothesis

Template-based analysis predicted a K-E63 salt bridge compensating the non-canonical P2 lysine (P2 Cα–Glu63 3.7 Å after minimisation;

Table 1. Model performance on MHCflurry-labelled test set.

Model	Acc (%)	Macro F1	WB F1	Params
MHCflurry†	100.0	1.000	1.000	–
Deep FFN	**91.9**	**0.921**	**0.880**	152,823
FFN (Jessen)	90.9	0.911	0.875	49,143
CNN	90.0	0.901	0.860	1,053,863
ResNet	84.3	0.847	0.792	324,931
LSTM	83.3	0.836	0.752	23,939
Random Forest	81.1	0.814	0.723	–

†MHCflurry evaluated on its own training distribution; 100% reflects label circularity.

Table 2. Deep FFN performance across labelling strategies.

Label source	Acc (%)	Macro F1	WB F1	CV mean (%)
PSSM (biophysics)	94.8	0.948	0.925	91.1 ± 0.9
MHCflurry 2.2.0	91.9	0.921	0.880	89.6 ± 0.8
Random synthetic	65.8	0.558	0.000	65.4 ± 0.4

Table 3. IEDB benchmark performance.

Metric	Value
Sensitivity	93.9% (46/49)
Specificity	75.0% (15/20)
Precision	90.2%
F1	0.920
ROC AUC	0.947

Table 4. Epitope-altering cancer hotspot mutations.

Mutation	WT peptide	Mutant peptide	Effect	Delta
p53 R248W	MNRRPILTI	MNWRPILTI	CREATED	+0.41
KRAS G12V	YKLVVVGAG	YKLVVVGAV	CREATED	+0.48
KRAS G12V	LVVVGAGGV	LVVVGAVGV	ENHANCED	+0.31
KRAS G12C	LVVVGAGGV	LVVVGACGV	ENHANCED	+0.31
KRAS G13D	GVGKSALTI	DVDKSALTI	DESTROYED	−0.47
KRAS A146T	GIPFIETSA	GIPFIETST	DESTROYED	−0.42
p53 R249S	GMNRRPILT	GMNRSPILT	DESTROYED	−0.16

Table 5. Non-canonical (Type B) sensitivity by predictor class.

Predictor	Canonical	Type B
Deep FFN (sequence)	0.979	0.607
MHCflurry (sequence)	0.986	0.668
MUNIS (structure)	0.950	0.647

−30 kcal/mol Coulombic estimate), presented as a testable structural hypothesis.

3.6 Multi-allele generalization

A single rebalanced pan DFFN reached non-canonical sensitivity (mean ± SD, 5 seeds): A*02:01 0.98 ± 0.02, A*01:01 0.92 ± 0.04, A*03:01 0.98 ± 0.02, B*07:02 0.92 ± 0.05, B*44:03 1.00 ± 0.00 — parity with per-allele models. On the 15-allele panel, pooled non-canonical sensitivity was 0.819 (12/15 alleles ≥0.76). Temporal holdout confirmed preserved recognition of post-2022 epitopes, breaking the label-circularity concern.

3.7 Non-canonical plateau

Canonical binders were recognized at ∼0.98 by every predictor class: DFFN 0.979, MHCflurry 0.986, MUNIS 0.950. Non-canonical (Type B) binders plateaued at ∼0.65: DFFN 0.607, MHCflurry 0.668, MUNIS 0.647 (Table 5) — structure-based prediction did not lift the ceiling. Type A binders were inherently hard (DFFN 0.477) and not recovered by broad-motif retraining. The extreme energy-compensated tail (P2-Lys/Arg/His reaching B-pocket Glu63) was scored as non-binder by all three predictors.

4 Discussion

Label provenance defines the achievable performance ceiling for peptide-MHC binding prediction; architecture choice is secondary. Across three labelling strategies, label source accounted for ∼29 pp of accuracy variation, dwarfing the ∼11 pp maximum inter-architecture gap, and the variation concentrated exclusively in the WB class (F1 0.723–0.925). This concentration is not incidental: the 50–500 nM affinity range defining the WB boundary is precisely where experimental binding probabilities are most graded, where ML predictors trained on finite data produce the most uncertain labels, and where conformational flexibility renders sequence-only approximations least reliable.

The three-tier label experiment — PSSM (94.8%) → MHCflurry (91.9%) → random (65.8%) — provides direct quantitative evidence for a label contamination cascade centred on the WB class. Deterministic biophysical rules (PSSM) yield the cleanest decision boundary because the model learns a ground-truth additive function. ML-derived labels (MHCflurry), themselves trained on IEDB data that is >55% computationally labelled (Preibisch *et al.*, 2026), already encode non-linear inter-position dependencies that reduce apparent performance; the 2.9 pp gap represents information loss from learning to approximate an approximation. If such predictions are re-deposited into public databases, each successive predictor generation compounds these biases. We propose that the WB F1 — a metric that collapses to zero when a model ignores the class — be reported as a standard diagnostic: a declining WB F1 across predictor generations would function as an early-warning signal for accelerating label contamination.

BLOSUM62 proved more robust than ESM-2 at single-allele training scales; high-dimensional per-position embeddings overfit small datasets. Architectural complexity beyond depth with batch normalisation gave diminishing returns, and a simple PSSM baseline (94.8%) exceeded all neural networks — underscoring that biophysical knowledge should be a standard comparator in computational benchmarking.

The multi-allele extension shows the recipe is portable: one rebalanced pan model replaces a per-allele panel across five alleles and reaches 0.819 pooled sensitivity on 15 alleles, approximating NetMHCpan's practical value with a simple DFFN. Yet non-canonical (motif-failing) binders plateau at ∼0.65 sensitivity across sequence- and structure-based predictors alike; the energy-compensated tail (P2-Lys/Arg/His reaching B-pocket Glu63) is prediction-resistant to all current methods. The three-tier screening pipeline we considered (sequence filtering → structural scoring → energy-minimisation validation) therefore remains a hypothesis for these configurations, pending experimental validation — consistent with the finding that the canonical binder problem is effectively solved while the non-canonical tail defines the current frontier.

4.1 Limitations

Training labels are MHCflurry-derived rather than experimental affinities; we quantified the circularity by cross-referencing the 49 IEDB epitopes against MHCflurry's training corpus — 44/49 (89.8%) overlapped, though all five non-overlapping epitopes were correctly classified. The single-allele design was deliberate — holding the allele constant isolates label-source variation — but the multi-allele extension addresses generalisability directly. The multi-allele estimates rest on IEDB-derived positives; experimental (non-IEDB) validation of the extreme tail is outstanding, and a pMHC-fine-tuned structure model might lift the Type B plateau. The K-E63 salt bridge, all translational predictions, and the cancer neoepitope candidates require experimental confirmation (e.g., IFN-γ ELISpot or tetramer staining). Antigen-processing steps (proteasomal cleavage, TAP transport) are not modelled.

5 Conclusion

Label provenance defines the performance ceiling for peptide-MHC binding prediction; architecture choice is secondary. The concentration of label-induced variation in the weak-binder class establishes it as a quantitative sentinel for label contamination, and we recommend that future benchmarks report WB F1 and label provenance as standard practice.

BLOSUM62 proved more robust than ESM-2 at single-allele training scales, and architectural complexity beyond depth with batch normalisation gave diminishing returns. A multi-allele extension showed that one rebalanced pan model replaces a per-allele panel across five common alleles (and reaches 0.819 pooled sensitivity on a 15-allele panel), while non-canonical binders plateau at $\sim$0.65 sensitivity across sequence- and structure-based predictors alike. The energy-compensated tail (P2-Lys/Arg/His reaching B-pocket Glu63) is prediction-resistant to all current methods, defining the current frontier. Experimental validation of the extreme tail is the decisive next step.

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